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1. What is a database? Define it in your own words and explains why databases are an essential in the modern world. Provide two real-world examples of where databases are used.

A database refers to a systematically arranged collection of data that is stored electronically, allowing for efficient access, management and easy updates. Instead of keeping information in a haphazard manner, databases organize data into tables, records and relationships. This is to facilitate the retrieval of valuable insights. They allow businesses or organizations to securely store information or data in vast amounts, it also ensures accuracy and helps in decision making.

For example:
- Banks utilize databases to monitor customer accounts and track their transactions in real time.
- Hospitals use databases to store patient records, track their treatments and manage referals and appointments. (IBM, 2023)

2. What is a Data warehouse? How does it differ from a regular database? ~~What~~ Why would a company use a data warehouse?

A data warehouse serves as an all in one storage space for extensive historical information gathered from various origins. In contrast to operational databases that manages everyday

transactions, data warehouses focus on examination and reporting. The main distinction lies in the fact that databases facilitate immediate operations, whereas data warehouses aid in making informed strategic choices.

For example: A retail business might utilize a data warehouse to scrutinize ~~a data~~ numerous years of sales records and recognize patterns such as a busy shopping period and consumer buying habits. (Microsoft, 2023)

3. What is a data lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

A data lake serves as a storage solution that accomodates unprocessed data in its native form, inclusive of structured, semi-structured and unstructured types. This enables companies to retail large quantities of data without the necessity to establish a structure beforehand.

A business may prefer a data lake when they have large volumes of unstructured or semi-structured data or when they want to store raw data whereas data warehouses are designed for structured data and query-focused analytics.

For example: A telecom company like Vodacom might use a data lake to store raw data from network logs, customer interactions and usage patterns. (AWS, 2023)

4. What is a data Mart? How does it relate to a data warehouse and who typically uses it?

A data mart is a mini version of a data warehouse, aimed specifically at a particular department like marketing, finance or sales. They enhance effectiveness by enabling teams to easily retrieve information that pertains to their needs.

For instance, a marketing group might utilize a data mart that holds data on customer demographics and the outcomes of campaigns to access their advertising tactics.

5. Explain the difference between structured, semi-structured and unstructured data. Provide one example of each.

Structured Data: This is a type of data that is clearly organized into rows and columns, each with a specific format.

Example: A table of customers at a bank featuring columns such as account number, name, balance and date of start. Each entry can easily be understood as it is straightforward.

Semi-Structured Data: This type of data is organized as well but allows for variability. It employs labels or keys to outline the fields, so entries are not required to be uniform.

Example: CSV files with inconsistent columns, email with headers and body.

Unstructured Data : This is a type of data that does not entail structure and lacks format. It consists of raw data / information that can not easily be organized.

Example : A recording or a video clip on social media.

Structured data	Semi-Structured Data	Unstructured Data
Ready for spread sheets	* Uses tags / keys * Adaptable fields	* Raw / unfiltered media, sound, & visuals

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe atleast two types of data models.

Data modeling refers to the method of organizing how information is arranged, saved and interconnected within a system. Serves as a frame work for constructing databases. It is important when designing a system that works with data because it guarantees that data remains consistent, it also minimizes duplication and enhances overall efficiency.

Logical models	Physical models
* Specific structure and connections. * Business view	* Real-world application in a database. * Technological view

7. What is Data mining? How is it used in business? Provide two real-world examples of data mining in action.

Data mining involves employing algorithms to identify

patterns, trends and insights from extensive datasets that may not be immediately apparent. Businesses or companies use data mining to forecast consumer actions, identify fraudulent activities and enhance marketing strategies. They also ~~the~~ also use it for easy informed decision making based on the compilation they get and not assumptions.

Examples: Retail Business (Choppies)

Analyzes sales recepts information to determine which items are frequently purchased altogether. Eg. Breakfast food are usually placed together in the same aisle if research show they are usually purchased at the same time.

Banking Business (FNB)

Analyzes transaction data to spot fraud. If a card is used in Botswana (Gaborone) and South Africa (Cape Town) within a span of ten minutes, the system spots this as unusual and halts the transaction.

8. What is a data management system (DBMS)? Name and briefly describe four popular DMS platforms (e.g MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

A database management system commonly known as a DBMS is a type of software that organises, handles and retrieves informations in a systematic manner.

MySQL = basic web

PostgreSQL = intricate SQL

Oracle = mostly for businesses

MongoDB = adaptable data

Platform	Suitable use
MySQL	* Relational Speedy * Highly utilized Web applications e.g WordPress
PostgreSQL	* Relational Robust functionalities * Powerful SQL Complicated queries
Oracle	* Relational designed for enterprises * Highly secure large organizations e.g biling systems.
MongoDB	* No SQL Adaptable, based on documents. * Raw data such as product listings in online stores.

9. What is ETL (Extract Transform Load)? Explain each step and describe why this process is important in data analytics?

ETL refers to a method of intergrating data, which involves gathering data from various origins, refining and organizing it, and then saving it in a centralized location such as a data warehouse for further analysis.

1. Extract

This step entails gathering unrefined data from several sources including databases, API's and spreadsheets.

2. Transform

During this phase the data is refined, arranged and formatted to make it functional. This may include eliminating duplicates or inaccuracies.

3. Load

In this phase, the prepared data is inserted into a specific system, typically a database or data warehouse.

NB In ~~suma~~ summary, ETL makes certain that ~~uf~~ unprocessed data is converted into a dependable and valuable resourse for analysis and informed business choices.

10. What are common data storage formats used in analytics (e.g CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

Common data storage formats utilized in analytics consist of CSV, JSON, Parquet and Avro, each tailored for various purposes.

CSV is a straightforward, easily readable format organized in rows, best for smaller datasets and sharing information.
JSON is a semi-structured format that can handle nested data, making it well-suited for API's and web applications.
PARQUET is a binary format organized by columns, optimized for extensive data analysis, providing excellent compression and quick query results when dealing with selected columns.
AVRO presents a row-based binary format with intergrated schema support, making it appropriate for data ~~support~~ streaming.

The decision on which format to use relies on aspects like the complexity of the data, its size and the performance needs.

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak) and General AI (Strong). Provide one example of each.

Artificial Intelligence (AI) - is a field withing Computer Science that aims to build machines or systems that can execute tasks typically requiring human intellect. These tasks encompass learning, reasoning, solving problems, perceiving and ~~dection~~ decision making.

1. Narrow AI - created to handle a particular task or a small set of tasks. E.g Siri - it is capable of answering questions, creating reminders and following voice commands but it can not think nor reason beyond specific abilities.

2. General AI - pertains to a system that can comprehend, learn and utilize knowledge across various tasks, akin to human intelligence. E.g Jarvis from Marvel movies - can handle numerous intricate tasks, grasp context and make choices like a person.

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised and Reinforcement Learning.

Machine Learning is a sector within AI that enables systems to identify trends from data, allowing them to make predictions or choices ~~wth~~ without needing direct instructions.

Difference between machine learning and traditional Programming:

TP: rules + data → result

ML: data + result → discovers rules (models)

Types of Machine Learning

1. Supervised Learning → Gains knowledge from data that has been labeled (e.g forecasting exam scores).
2. Unsupervised Learning → Detects trends in data that lacks labels (e.g clustering customers).
3. Reinforcement Learning → Acquires ~~knote~~ knowledge through a trial and error method using rewards and penalties (e.g AI for gaming).

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neutral Networks and how do they work?

DL is a branch of Machine Learning ~~that~~ employs layered models known as neutral networks to learn patterns from extensive datasets autonomously, particularly for applications such as image, voice and text identification

Neutral Networks

These are digital models that draw inspiration from the human brain and consist of layers of interconnected 'neurons' (input, hidden, output). This network enhances its ability to provide accurate predictions or choices.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you we in everyday life.

NLP is a branch of ~~artificial~~ artificial intelligence that enables machines to comprehend, analyze and produce human language.

3 daily uses: ① Text correction /predictive typing

* Our smartphones keyboards correcting mistakes and offering word suggestions.

② Voice-controlled helpers

* Instructing siri or google assistant to 'call mum' or 'wake me up at 7am'.

③ Unwanted mail detection

* Gmail identifying and sorting spam based on the content of the emails or messages.

15. What is ~~A~~ Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLM's and explain what they are used for.

LLM is an AI system developed using extensive text collections to grasp and produce human language. It forecasts the next word or token based on the surrounding context. An LLM is different from a standard ML model through scale: LLM's contain billions of parameters whereas tradition ML models only have thousands or millions. They also differ through training: LLM's are trained through self-supervised learning on textual data while standard ~~E~~ ML models usually require labeled information for a single task.

TYPES OF LLMS

1. GPT-4 - Employed for writing assistance, education e.g CHAT GPT.
2. Gemini - Utilized for search queries, summarization e.g Google's AI.

NB: Standard ML is like a specialist, whereas LLMs are more like generalists in the realm of language.

16. What is generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

Generative AI refers to AI that produces original content such as written material, picture or programming code. It analyzes extensive data collections to understand patterns, subsequently creating new outputs by anticipating what aligns with the given prompt.

TOOLS THAT USE GENERATIVE AI

① CHATGPT - creates written content, provide answers to inquiries and clarifies programming code.

② Midjourney - Produces visuals based on textual descriptions

③ Github Copilot - Proposes coding suggestions while you write programs.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

TD - The information utilized to instruct an AI model by presenting it with samples of inputs and their appropriate results. The quality is important because the model identifies trends based on this information. Poor quality data leads to flawed patterns and consequently inaccurate predictions. If data is biased/incorrect the model produces unfair, inaccurate or harmful outcomes. E.g an AI used for hiring and biased data is provided will lead ~~on~~ to dismissal of qualified candidates.

Training Dataset

18. What is computer vision? How do machines 'see' and interprete images? Provide two real-world use cases of computer vsion.

CV refers to technology enabling machines to interprete images and videos. Machines 'see' and interpret through visuals which are transformed into pixels and numerical data. Systems recognize patterns such as contours, forms and hues to pinpoint objects.

Examples

① Face unlock / recognition - The camera on your phone recognizes your facial features to enable access or to unlock the device.
② Self checkout - Cameras in stores read products without barcodes for automatic biling.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are API's important in modern software and data systems?

API is a collection of guidelines that enables various software systems to communicate with one another. It resembles a server in a dining establishment. You inform the server of your order, they relay it to the kitchen and they return with your meal (simple analogy).

API's are vital in moden software and data systems because they allow applications to exchange information and functionalities without the need to recreate everything from scratch. E.g Uber utilizes the Google maps API for navigation instead of creating its own mapping system.

20. What is an API key? What is its purpose? How should it be stored or protected, and what could happen if it is exposed publicly?

An API key is a distinct ~~inde~~ identifier that verifies and authorizes an application or individual to access an API.

It confirms the identify of the requester and regulates entry and usage constraints.

How to protect: Keep it in environment variables or secret storage, never in source code or version control. Restrict permissions and change it regularly.

If exposed publicly, malicious actors can misuse your API, generate excessive costs, take sensetive information or mimic your application.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

REST - Representational State Transfer
It is an API that follows REST architecture using HTTP to access and manipulate resources.

① GET - ~~Rect~~ Read data e.g customers returns customer list.
② POST - Create new data e.g send new customer details.
③ PUT - Update/replace exuting data e.g replacing all fields for customers.
④ DELETE - Remove data e.g delete customer A or customer B.

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

JSON: A lightweight text format for storing and sending data using key-value pairs.

APIs use it because it is human readable, easy for machines to parse, and language-agnostic. Standard for web APIs.

example: Keys on the left, values on the right

order id : 5021
customer : Dapill Kefentse
item(s) : iPad
total : 11 999.99
paid : true

APIs pass data like this because both systems can read it easily.

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like CHATGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

PE - Crafting input prompts to get useful, accurate output from AI models. It is an important skill when working with AI tools because AI outputs ~~prompts~~ quality depends on input clarity. Good prompts reduce hallucinations, save time and return data in the format you need.

Examples of poorly written prompts

① Poor - "Tell me about sales."
Improved - "Summarise Q1 2026 sales trends for our Maun store."

② Poor - "Make a graph."
Improved - "Act as a data analyst. Generate ~~pla~~ python code using matplotlib to plot monthly revenue for 2025 as a line chart with labels and title."

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or Javascript.

MAP function - Applies a function to every item in a list / collection and returns a new list with the results. Used to transform data without loops. Example:

Input list : [2, 4, 6]
Operation : "multiply each by 3"
Output list : [6, 12, 18]

MAP takes every item, does the same rule to it and gives you a new list.

25. What is cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Cloud Computing: Utilizing online servers for data storage, handling, and software instead of relying on local devices.

Service models : IaaS , PaaS, SaaS

① IaaS - Infrastructure as a Service
Offers basic computing elements such as servers and storage on a rental basis. You take care of the operating system and applications Example : AWS EC2.

② PaaS - Platform as a Service
Supplies a development environment along with tools to create and launch applications. Your focus is on code, not server maintenance. Example : Heroku.

③ SaaS - Software as a Service
Access software directly through the web without the need for an installation. Example : Google Workspace.

26. What is a Version Control System? What is Git and why is it ~~we~~ widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

VCS - Tracks changes to files over time so you can view history, revert and collaborate without overwriting work.

Git - A distributed version control system. Widely used because it is fast, works offline, handles branching well, and it is the standard for code / data projects on platforms like Github.

Commit : A saved snapshot of your changes with a message describing what changed.

Branch : A separate line of development to work on features or fixes without affecting the main codebase.

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data scientists?

JN - A web application that allows users to create and execute code within separate sections, intergrating written content, graphs and results into a single file. The most common language is 'Python'.

Data analysts and scientists regard JN popular because it intergrates programming, images and descriptions all in one spot. It also facilitates exploratory data analysis, model testing and sharing findings in a sequential manner.

28. What is Python? ~~Wha~~ Why is it considered one of the most popular programming languages in Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Python - A programming language that is high-level and easy to understand featuring straight-forward syntax.

It is favoured in data analytics because it carries a vast array of libraries, beginner-friendly, a robust community and effectively manages data, statistics and machine learning.

Frequently used libraries

* Pandas - Used for manipulating and analyzing data through DataFrames.
* NumPy - Efficient operations for numbers and mathematical arrays.
* matplotlib - Provides simple plotting capabilities and graphical representations.
* scikit-learn - Offers tools for machine learning models and preprocessing data.

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which ~~EEA~~ countries it applies to.

DP - The protection of personal data from unauthorized access, use or sharing. It prevents identity theft, maintains trust, and ensures legal compliance when handling sensetive information.

GDPR - EU law that sets strict rules for collecting / processing personal data. Gives individuals rights like access, deletion and consent.

* It applies to all 27 EU (European Union) countries plus EEA states (Norway, Iceland and Liechtenstein.) and it also applies to any organisation handling EU residents' data globally.

30. What is Data Ethics? Provide three examples of unethical use of data or AI and explain what makes each example problematic.

DE - Guidelines that promote just, clear and accountable practices for gathering and utilizing data.

Unethical uses of data

① Biased Hiring AI - Trained on male-dominated data, rejects women.
Problem - Reinforces discrimination and promotes gender based discrimination

② Deepfake videos - Videos of real people without their consent / permission.
Problem - Misinformation and reputational harm.

③ Selling user data without consent - App shares health data with adverts.
Problem - Violates privacy and trust.

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

DG refers to guidelines and procedures that guarantee data remains precise, safe and appropriately utilized throughout an organization.

Key components of good data governance framework

① Roles and ownership - designated personnel accountable for datasets.
② Data quality - criteria for precision, thoroughness and uniformity.
③ Security & privacy - measures for access control and regulations like GDPR.
④ Metadata & cataloging - Documentation that explains meaning and location of data.

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences

Bias in AI: when a system generates unequal outcomes due to the influence of human or systematic biases in the data it was trained on.

Example: The COMPAS risk assesment used in American courts rated Black defendants are more likely to reoffend compared to white defendants with the same background. (racism).

Consequences: Incorrect rejections of bail or parole, perpetuated discrimination and diminished confidence in the legal system.

33. What are the key differences between a Data Analyst, a Data Scientist and a Data Engineer and an AI Engineer? Create a simple table comparing their roles, skills and tools.

Role	Main Role	Key Skills	Common Tools
① Data Analyst	* Analyze data to answer business questions and create reports.	* SQL, statistics, data viz, Excel	* Excel, SQL, Pandas, Tableau, Power BI.
② Data Scientist	* Build models to predict trends and find patterns.	* Statistics, ML, programming experimentation.	* Python, R, Jupyter, Scikit learn, ~~Snowflake~~, SQL.
③ Data Engineer	* Build / maintain pipelines to move and store data.	* ETL, databases, cloud & coding.	* Python, SQL, Spark, Airflow, AWS, Snowflake.
④ AI Engineer	* Deploy and optimize AI/ML models in production.	* ML operations, software engineering, deep learning, APIs	* Python, PyTorch, TensorFlow, Docker, ML Flow.

Key Difference:
- Analysts explain what happened.
- Scientists predict what will happen.
- Engineers build data infrastructure.
- AI Engineers make models run in real products.

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g Power BI, Tableau) and explain how they help businesses make data-driven decisions.

BI - Turning raw business data into dashboards, reports and ~~ts~~ insights for decision-making.

① Power BI

* Connects to data resources and makes interactive dashboards.
* Helps track KPIs (key performance indicators) and spot trends fast.

② Tableau

* Visualizes large datasets with drag and drop charts ~~the~~
* Helps non-technical users explore data and find patterns.

PERSONAL REFLECTION

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

* Artificial ~~Bias (AB)~~ Intelligence Bias (AI Bias)

I was actually surprised how models can repeat human prejudice from training data, like COMPAS overestimating risk for some groups. It showed me why ethics matters in analytics.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

*Data governance - The roles, policies and standards felt abstract at first. I used YouTube tutorials, CHATGPT, Meta AI to map out how ownership and data quality connect.

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

*As an aspiring data analyst, I will use BI tools like Power BI to build KPI dashboards for business decisions. In daily life, AI already affects me through recommendation systems on SHEIN, Netflix and Showmax and fraud alerts from my bank (FNB) and network provide (MASCOM).

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